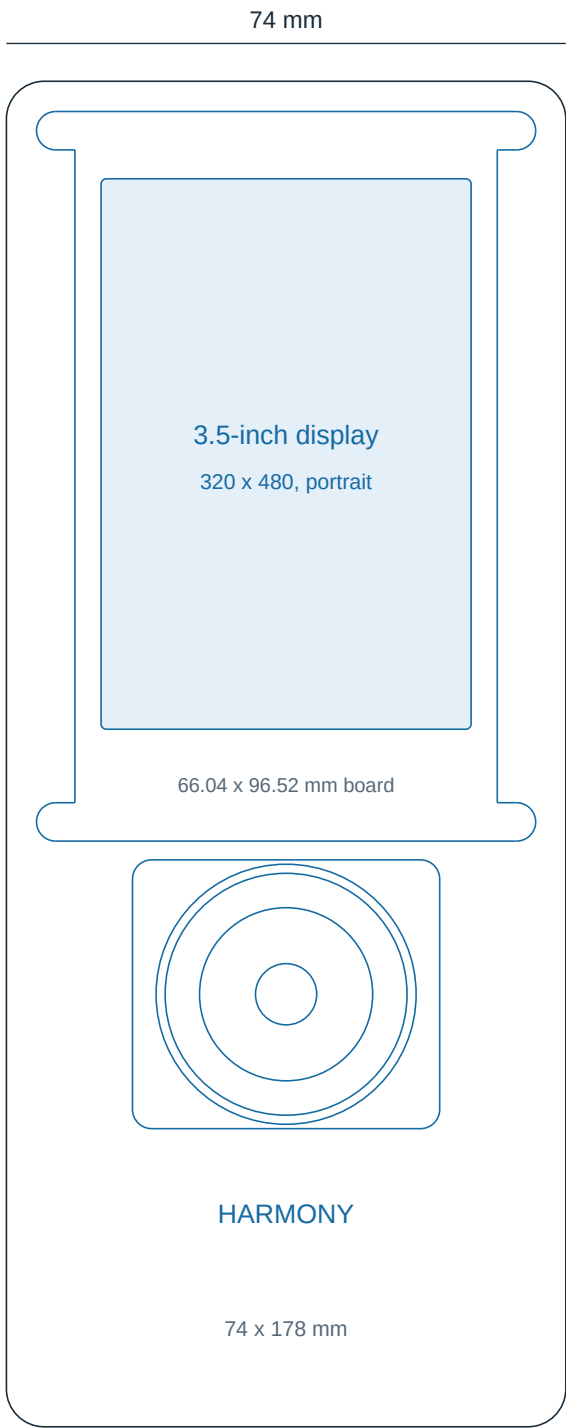


Smaller layout: full-size hand check

3.5-inch display, rotated ANO controls, and the selected factory-headered Feather.



12 mm SHORTER

Previous complete study: 190 mm.
This arrangement: 178 mm.
Width remains 74 mm.
Still a large player; try it in hand.

WHAT CHANGED

ANO board rotated 90 degrees.
Antenna location taken from CAD.
Feather moved closer to controls.
USB connector faces the left side.

PRINTABLE SIZE DUMMY

74 x 178 x 28 mm overall.
Print STL at 100%, flat back down.
Screen and wheel are recessed cues.
It has no electronics cavity.
Depth and button feel are assumed.

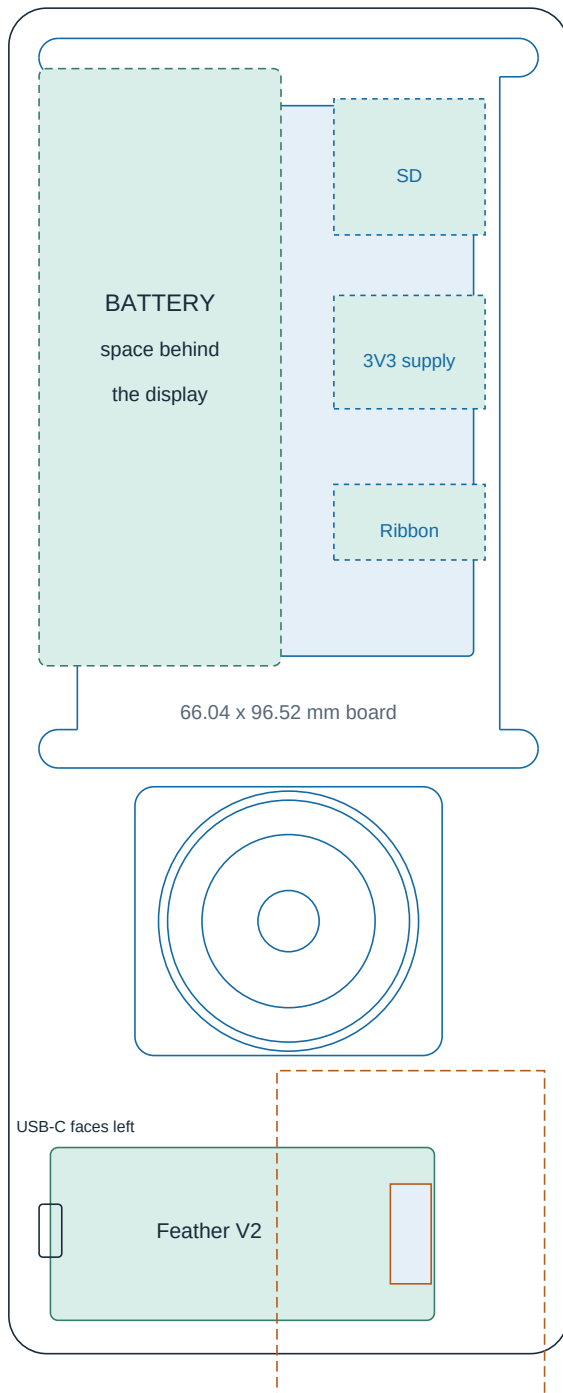
CHECK BEFORE ROUTING

Pocket fit and thumb reach.
Real socket and display heights.
Cable bends and plug clearances.
Carrier antenna cutout and RF.

Paper and solid dummy check size only. Neither is a finished electronics enclosure.

Planar placement and radio reservation

Front view; rear reservations overlap the display in projection. Z-stack is not validated.



74 x 178 mm proposal

Feather board: 50.8 x 22.86 mm.

Bottom margin: 4.41 mm.

Controls-to-orange-zone gap: 2 mm.

ORANGE: 15 mm HALO

Based on vendor antenna rectangle:

5.4 x 13.2 mm on the Feather.

Other module boxes stay outside.

Feather socket pads enter this zone.

Carrier cutout/copper review remains.

THIS IS NOT AN RF PASS

Halo extends 5.76 mm below shell.

Plastic and hand effects untested.

The full 3-D clearance is unverified.

Final range/throughput tests remain.

CABLE AND BATTERY SPACE

Use upper ANO QT connector;

plug/bend volume still unmodeled.

Reserve 32 x 79 x 10.5 mm for pack.

USB first; no battery connection.



Espressif recommends antenna overhang/clearance and testing in the final housing. See page 3.

Dimensions, changes and limits

Placement B supersedes the earlier 74 x 190 mm conservative arrangement.

Display board	66.04 x 96.52 mm
Adafruit #5846 Eagle board outline	
Display active-area guide	48.96 x 72.82 mm
Vendor package drawing; aperture not finalized	
ANO board, rotated	40.64 x 35.56 mm
Same board turned 90 degrees; firmware directions must follow	
ANO outer wheel guide	34.4 mm diameter
ENCODER_ANO vendor package circle; printed cue only	
Feather board	50.8 x 22.86 mm
USB adds 1.5 mm on the left; port opening not designed	
Antenna restrict rectangle	5.4 x 13.2 mm
Vendor X3 package, layer 41, transformed by R270	
GlobTek battery body	31 x 78 x 9.3 mm
Reserved 32 x 79 x 10.5 mm; swelling allowance unqualified	
Complete size dummy	74 x 178 x 28 mm
Proposed width/height; depth is provisional	
Earlier complete study	74 x 190 mm
12 mm longer; oversized antenna proxy replaced	

Source CAD and hashes: [carrier-rev-a/reference/design-sources.json](#).

RF reference: Espressif ESP32 Hardware Design Guidelines, PCB Layout Design.

Planar checks do not validate assembly depth, sockets, ribbon bends, thermal behavior or RF.

Carrier assembly still includes all sockets/connectors. No hand soldering is added.

Do not use this study as a drill, wiring, battery-harness or PCB manufacturing drawing.